

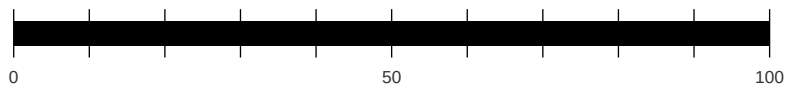
AprilTag reprint - CYLINDER tag id 0 at 40.0 mm

PRINT AT 100% / "Actual size". Turn OFF "Fit to page" and "Shrink oversized pages" - they scale silently.

Use a LASER printer. Inkjet black dye is often transparent at 850 nm - invisible to the D435 infrared imagers.

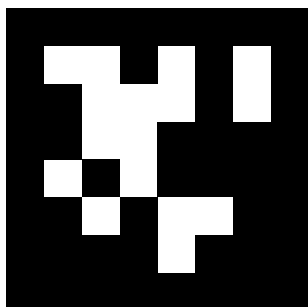
Check the bar below with calipers BEFORE cutting, then measure a printed tag and pass THAT number as tag_size.

Replaces the 30 mm vane tag, which stopped decoding partway through 5 of 12 runs while the 40 mm reference tag never did.

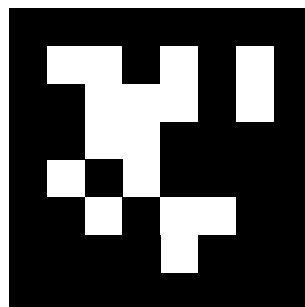


100.0 mm exactly. Short = your printer scaled it.

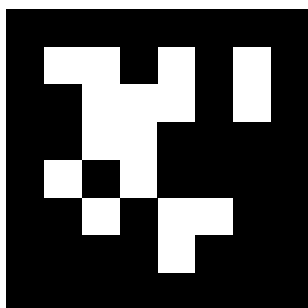
Four identical copies - three are spares. Solid black square = 40.0 mm. White patch incl. quiet zone = 60.0 mm.



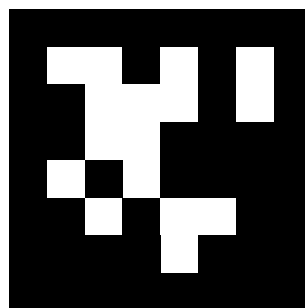
tag36h11 id 0 tag_size 40.0 mm



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MOUNTING. Keep the tag centre at 71.0 mm along the shaft axis (CYL_TAG_AXIAL_MM) and the vane normal along the pinch axis, so the turn stays an in-plane rotation. Orientation matters as much as position: the detector's pose frame is fixed to the pattern, so mount this square the same way up as the one it replaces. The patch is 60 mm across, 15 mm wider than the 30 mm tag's - check vane clearance.

Then set CYL_TAG_SIZE_M = 0.040 in `src/morphohand/bench/tags.py`. A tag printed at 40 mm and read as 30 mm misplaces the object by 33%.